

Full-Graph versus Specialty-Subgraph Features for Drug–Drug Interaction Type Prediction

An Inductive, Hypothesis-Driven Evaluation Across
Random Forest, GCN, and Retrieval-Augmented LLMs

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Abstract

Drug–drug interaction (DDI) prediction systems are increasingly deployed in specialty clinical contexts, but it is unclear whether subgraphs restricted to a specialty (e.g. oncology) capture sufficient structure for accurate type prediction, or whether full-network features are required. We test the hypothesis that graph features computed over the full DrugBank-derived interaction network (1,701 drugs, 191,541 typed directed interactions, 7 interaction types) yield significantly better DDI type prediction than features computed over an oncology subgraph (97 drugs, ~22K edges). We construct four scopes (full, oncology, liquid cancer, solid cancer), compute 14 graph features (10 node-level, 4 pairwise) under a strict inductive protocol that removes test edges before feature computation, and compare a feature-based Random Forest (RF) to a 2-layer Graph Convolutional Network (GCN). RF macro-F1 declines monotonically with scope (full 0.926, oncology 0.848, liquid 0.727, solid 0.710). McNemar’s test on shared test pairs — confirmed by permutation tests throughout — establishes Full > Oncology ($p = 0.036$) and Full > Liquid ($p = 0.0003$). Full vs Solid is non-significant on shared pairs ($p = 0.901$) and is explained by rare-class enrichment in Solid’s full test set; Liquid vs Solid ($p = 0.504$, $n = 23$) is underpowered. The GCN is non-monotone in scope (0.36–0.47), indicating that the scope effect is mechanism-specific to models that consume explicit pairwise features. A separate LLM follow-up finds that GPT-4o-mini reaches macro-F1 0.219 zero-shot and 0.232 with raw graph features supplied as text, but 0.839 with $K = 5$ retrieved labeled analogues, suggesting representation rather than information is the LLM bottleneck.

1 Introduction

Drug–drug interactions (DDIs) are a major source of preventable harm in clinical practice. Computational DDI prediction has matured rapidly, with modern systems using molecular structure, knowledge graphs, and large language models to predict whether two drugs interact and, in the finer-grained variant we study here, what *type* of interaction they will produce [1, 2, 3].

Most published DDI predictors are trained on the full DrugBank network [4] or comparable broad-coverage resources. Yet many of the most pressing clinical use cases are specialty-bounded: an oncology pharmacist managing chemotherapy regimens does not, in practice, reason over the same drug universe as a cardiologist managing anticoagulation. This raises a methodological question that is often left implicit: are full-network graph features actually necessary when the deployment context is a specialty subgraph, or could a specialty-restricted model do as well?

DDI prediction is naturally a graph problem. Each drug is a node, each documented interaction is a directed edge, and prediction therefore depends not only on individual drug-pair properties but also on the structural context in which a pair sits. From this graph framing two nested questions emerge:

- **Q1.** Do graph-structural features help predict DDI type at all?
- **Q2.** If they do, does the *scope* of the graph matter — does restricting to a clinically focused domain help or hurt?

This paper focuses on **Q2** as a falsifiable hypothesis test:

H₁. *Graph features computed over the full DDI network yield significantly better DDI type prediction than features computed over an oncology-only subgraph.*

The treatment is full-network features; the baseline is oncology-only features; the metric is macro-F1 on held-out drug pairs; the test is McNemar’s test on shared correctness vectors, with permutation tests providing a

parallel non-parametric check. We additionally compute features over two oncology-internal scopes (liquid cancers, solid cancers) to probe a dose-response, and we run the same comparison through a feature-based Random Forest and a graph-native GCN to probe whether the scope effect is a property of the data or of the model.

Contributions. This paper contributes:

1. A clean inductive evaluation of graph-scope effects in DDI type prediction, with statistical tests on shared test pairs that decouple model-comparison signal from sampling differences across scope-specific test sets.
2. Evidence that the RF macro-F1 advantage of full-graph features is real but partly observational: label distributions differ systematically between scopes, so the scope effect is confounded with label mix.
3. The finding that a 2-layer GCN is scope-insensitive while RF is scope-sensitive, identifying the scope effect as mechanism-specific to models that consume explicit pairwise features — pinned by feature-importance analysis to common neighbors and Jaccard similarity.
4. Evidence from a separate LLM follow-up (Section 7) that the LLM bottleneck is representation, not information: raw features supplied as text barely improve over zero-shot, but $K=5$ retrieved labeled analogues lift macro-F1 to 0.839.

2 Related work

2.1 DDI prediction

Computational DDI prediction has progressed from molecular-similarity methods that score pairs by chemical similarity to known interacting pairs [1, 2], through knowledge-graph approaches that embed drugs in a multi-relational network [3, 5], to recent work using language models for textual interaction descriptions. Most of these systems are trained and evaluated on the full DrugBank network [4] or comparable broad-coverage resources, with limited examination of whether the network scope is appropriate for the deployment context. The question we test in this paper — whether full-network features remain advantageous when the deployment context is a specialty — has not, to our knowledge, been isolated as a falsifiable hypothesis in prior DDI literature.

2.2 Graph features for biological networks

Classical pairwise features for link prediction — common neighbors, Jaccard similarity, Adamic-Adar, resource allocation — were introduced for social-network link prediction [6] and have been widely applied in biological networks under the umbrella of *network medicine* [7, 8]. The underlying intuition is that shared

neighborhood encodes shared biological function: drugs sharing many known interaction partners likely share targets, metabolic pathways, or pharmacological mechanisms, and their unobserved interactions can be inferred from that overlap. The informativeness of these features is, by construction, sensitive to neighborhood density — a fact our experiments will exploit when interpreting the scope effect.

2.3 Scope effects in graph ML

Graph convolutional networks [9] replace explicit pairwise features with learned representations propagated through message passing. Inductive variants such as GraphSAGE [10] were introduced precisely to handle the train/test edge separation we adopt in Section 4.3. The *over-smoothing* phenomenon — that deep or dense message passing collapses node representations toward a common mean and degrades class separability [11] — is directly relevant for our dense ($\bar{d} \approx 225$) DDI graph. Beyond GCNs, the broader question of how subgraph selection affects downstream performance has been studied for general graph learning but remains underexplored in DDI specifically; our work is one attempt to close that gap with a controlled comparison.

3 Data and graph construction

3.1 DDI corpus

We use a publicly available Kaggle DDI dataset derived from DrugBank [4], comprising 191,541 typed pairwise interactions over 1,701 drugs, treated as a directed graph (the source drug is the one whose pharmacological effect is modulated by the object drug, where direction is identifiable from the description template). Each interaction is annotated with a free-text description; we extract one of seven interaction types from these descriptions using a regex pipeline that matches against 89 semi-structured description templates curated from the corpus. The resulting type label has 99.997% coverage (5 unparsed records dropped). A representative example: “*The metabolism of Digoxin can be increased when combined with Rifampicin.*” → **metabolism**.

The seven interaction types are, in order of corpus frequency: *adverse_effects* (32%), *activity* (24%), *metabolism* (21%), *concentration* (18%), *efficacy* (4%), *excretion* (1%), and *absorption* (0.5%). The label distribution is therefore heavily imbalanced, dominated by four mid-to-high-frequency classes and three rare classes. We use macro-F1 as the primary metric throughout because it gives equal weight to each class and is therefore informative about rare-class behavior.

The full graph has average degree ~ 225 , indicating a very dense interaction landscape. We treat this density as a property of the data rather than a defect: it is precisely the regime in which structural pairwise features should carry signal, and it is also the regime

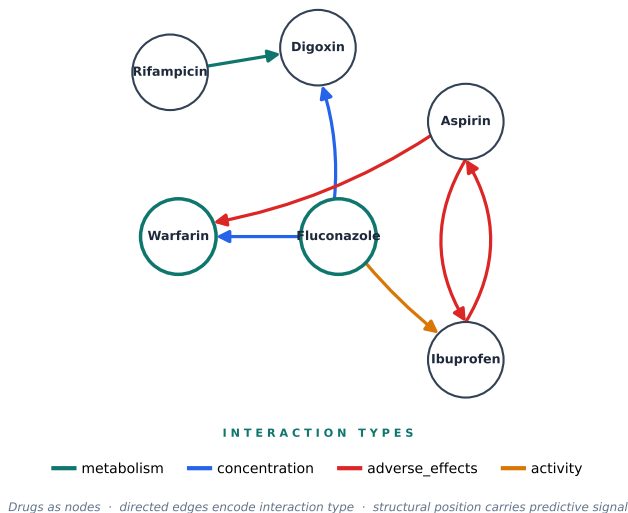


Figure 1: Schematic of the DDI graph. Drugs are nodes; directed edges encode interaction type. The example shown captures four of the seven types we predict: *metabolism* (Rifampicin → Digoxin, increased metabolism), *concentration* (Fluconazole modulating Warfarin and Digoxin), *adverse_effects* (Aspirin-Ibuprofen, Aspirin-Warfarin), and *activity* (Fluconazole → Ibuprofen). Structural position in this graph carries the predictive signal our methods exploit.

in which over-smoothing in graph neural networks becomes a salient concern (see Section 4.2).

3.2 Specialty subsets

We construct three specialty subsets of DrugBank drugs based on therapeutic-area annotations:

- *Oncology* (97 drugs): all drugs flagged as antineoplastic or used in cancer therapy.
- *Liquid cancer* (35 drugs): drugs targeting hematological malignancies (leukemias, lymphomas, multiple myeloma) — e.g. imatinib, ibrutinib, venetoclax, bortezomib.
- *Solid cancer* (62 drugs): drugs targeting solid tumors — e.g. erlotinib, vemurafenib, cisplatin, tamoxifen.

Together, the liquid and solid subsets approximately cover the oncology set with minimal overlap.

The subgraph G_S for drug set S retains every edge (u, v) such that $u \in S$ or $v \in S$ — that is, an edge is kept whenever at least one endpoint belongs to the specialty. The resulting subgraphs are therefore *domain-focused slices* rather than strict induced subgraphs.

Two reasons motivate the inclusive rule. First, the strict rule $(u \in S \wedge v \in S)$ produces graphs too small for stable feature estimation in the liquid and solid subsets. Second, the inclusive rule reflects the realistic deployment scenario in which a specialty DDI system reasons about drugs *touching* the specialty, including cross-specialty pairs. The resulting subgraph sizes are: oncology $\sim 22,000$ edges, liquid $\sim 8,300$ edges, solid $\sim 14,400$

edges. Figure 2 visualizes the four scopes side-by-side.

4 Methods

4.1 Random Forest with explicit graph features

The main model is a Random Forest classifier over hand-crafted graph features, chosen for its interpretability and its ability to expose which features drive predictions across scopes.

We compute 14 features per directed drug pair (u, v) , partitioned into 10 node-level features (5 per endpoint) and 4 pairwise features.

Node-level features (5 per endpoint, 10 total).

- *In-degree* and *out-degree* — the number of incoming and outgoing typed edges.
- *Betweenness centrality* — fraction of shortest paths through the node.
- *Clustering coefficient* — local triadic density.
- *PageRank* — global structural centrality under a random-walk-with-restart model.

Pairwise features (4 total).

- *Common neighbors* $|N(u) \cap N(v)|$, where $N(\cdot)$ is the neighbor set in the underlying undirected graph.
- *Jaccard coefficient* $|N(u) \cap N(v)| / |N(u) \cup N(v)|$.
- *Same-community indicator* — 1 if u and v fall in the same Louvain community [12] on the train graph, else 0.
- *Degree difference* $|d(u) - d(v)|$.

These features are intentionally classical and interpretable. We chose not to use learned embeddings or external molecular features at this stage: the goal is to test whether a scope-vs-feature relationship is visible at the most transparent end of the graph-feature spectrum, where individual feature contributions can be audited and connected to network-science intuition.

Classifier. We train scikit-learn’s `RandomForestClassifier` with 500 trees, `max_depth` tuned on a held-out validation slice of the full-scope training set, balanced class weights to mitigate label imbalance, and a fixed random seed for reproducibility. Hyperparameters were selected on the full graph and *frozen* across scopes to avoid scope-conditional overfitting.

4.2 2-layer GCN

As a graph-native alternative we train a 2-layer Graph Convolutional Network [9]:

$$\begin{aligned} & \text{5-dim node feat.} \rightarrow \text{GCN}_1(64) \rightarrow \text{GCN}_2(64) \\ & \rightarrow \text{concat}(h_u \parallel h_v) \rightarrow \text{MLP} \rightarrow \text{7-way softmax.} \end{aligned}$$

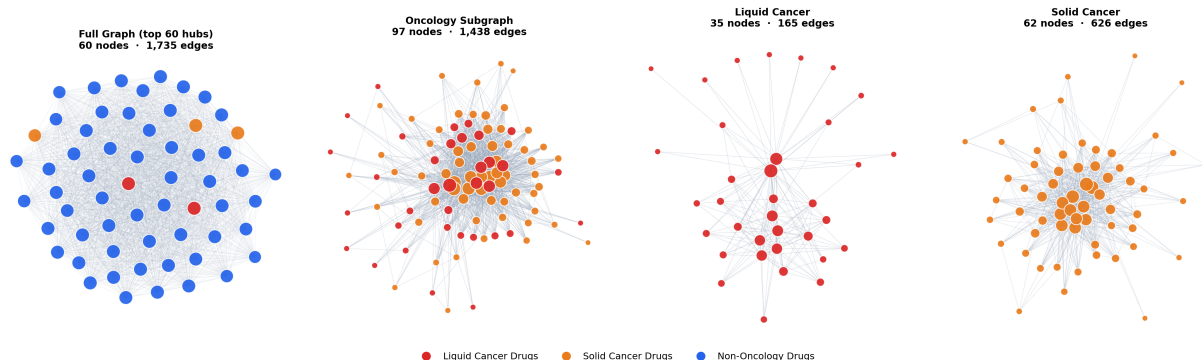


Figure 2: The four graph scopes, visualized as induced subgraphs (edges retained only when both endpoints are in the target drug set). For the full graph panel, the 60 highest-degree drugs are shown since the full network is unrenderable as a node-link diagram. Node color encodes drug category (**liquid cancer**, **solid cancer**, **non-oncology**); node size is proportional to degree. *Visualization uses induced subgraphs for legibility; the analysis in Sections 4–5 uses the inclusive subgraph rule defined in Section 3.*

Each node is initialized with a 5-dimensional feature vector containing the same node-level descriptors used by RF (in-degree, out-degree, betweenness, clustering, PageRank). The two convolutional layers each output 64-dimensional embeddings; endpoint embeddings are concatenated to a 128-dimensional pair representation and passed through an MLP head producing logits over the seven types. The model is trained with cross-entropy and Adam, and early-stopped on validation macro-F1.

A predictable concern for this baseline is *over-smoothing*: with average degree ~ 225 , repeated graph convolution mixes node representations across neighborhoods aggressively, which can degrade class separability [11]. The GCN therefore serves both as the end-to-end counterpart to the RF + explicit-features approach and as a way to test whether the scope effect is automatically captured by any graph model, or whether it depends on having explicit pairwise features in the input.

4.3 Inductive evaluation protocol

A central methodological choice in this study concerns *when* test edges are removed from the graph. In a transductive setup, all edges (including held-out test edges) participate in feature computation; the model is then asked to predict the type of a test edge whose endpoints’ graph features were partly defined by the test edge itself. In our pilot RF, this transductive setup yielded macro-F1 ≈ 0.97 — a result we ultimately judged untrustworthy due to feature leakage, particularly through neighborhood-overlap features that directly encode whether a candidate edge already exists.

We therefore adopt an inductive protocol throughout this paper. For each scope S and each train/test split, we:

1. Stratify-split the typed edges into 80/20 train/test, stratifying on interaction type to preserve rare-class

proportions.

2. **Remove test edges from the graph** before any feature computation.
3. Compute all 14 graph features on the train-only graph.
4. Train RF and GCN on train edges; evaluate on test edges using features computed in step 3.

The shift from transductive to inductive reduced full-scope macro-F1 from approximately 0.97 to 0.926 — substantial, but still strong, and now defensible. We treat 0.926 as the headline number for full-scope RF throughout the paper. The inductive control is, in our view, the single most important design choice in this study: any scope effect we report must be a scope effect on a model that has not seen its test edges before features are computed.

4.4 Statistical tests

The primary metric is macro-F1 across the seven interaction types. We report 95% confidence intervals via 1,000-sample bootstrap over test edges [14]. For pairwise model and scope comparisons, we use **McNemar’s test** [13] on shared test pairs — that is, drug pairs that appear in the test sets of both scopes being compared — comparing per-pair correctness vectors. This isolates the model-comparison signal from differences in test-set composition that would otherwise contaminate raw macro-F1 comparisons across scopes.

As a parallel non-parametric check we run **permutation tests** on the same shared-pair correctness vectors with 10,000 label permutations. Permutation p -values agree with McNemar’s at every comparison reported in Section 5.2, so we report McNemar p -values in the main text.

Scope	Drugs	Edges	macro-F1	95% CI
Full	1,701	191,541	0.926	[0.921, 0.931]
Oncology	97	~22K	0.848	[0.799, 0.878]
Liquid	35	~8.3K	0.727	[0.703, 0.744]
Solid	62	~14.4K	0.710	[0.678, 0.845]

Table 1: Inductive RF macro-F1 across the four scopes, with 95% bootstrap CIs over test edges. Random baseline ($1/7$) = 0.143.

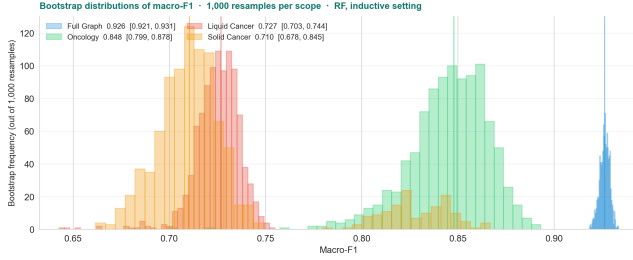


Figure 3: 1,000-resample bootstrap distributions of test-set macro-F1 for the four scopes. Tight peaks indicate stable estimates; wide bells indicate rare-class instability.

5 Results

5.1 Main scope effect

Table 1 reports inductive RF macro-F1 with 95% bootstrap CIs across the four scopes.

The pattern is monotone in scope size, consistent with H_1 ’s directional prediction: $0.926 > 0.848 > 0.727 > 0.710$. Solid’s CI is notably wider than its companions, reflecting rare-class instability under bootstrap resampling at smaller scope: Solid’s test set contains only a handful of edges in some of the rarer types, and re-samples that miss those edges entirely produce sharply different per-class F1.

5.2 Statistical validation

Bootstrap CIs around scope-level macro-F1 are influenced both by model performance and by test-set composition (each scope has its own test set, with its own type distribution). Figure 3 visualizes the per-scope bootstrap distributions: full-graph and oncology distributions are clearly separated, while liquid and solid distributions overlap heavily — visually anticipating the underpowered statistical comparison reported below. Solid’s distribution is also markedly wider than the others, consistent with rare-class instability.

To isolate the model-comparison signal from test-set composition, we run McNemar’s test on the *shared* test pairs of each scope pair; permutation tests confirm every conclusion below.

The Full > Oncology and Full > Liquid contrasts reach significance at conventional thresholds. Two comparisons need careful interpretation:

Comparison	n_{shared}	Δ acc.	p	Sig.
Full vs Oncology	888	+0.025	0.036	*
Full vs Liquid	323	+0.081	0.0003	***
Full vs Solid	623	+0.003	0.901	n.s.
Liquid vs Solid	23	-0.130	0.504	n.s. (UP)

Table 2: McNemar’s test on shared test pairs. Δ accuracy is pairwise accuracy difference on the shared subset only. UP = underpowered. Significance: * $p < 0.05$, *** $p < 0.001$.

Scope	GCN macro-F1
Full	0.44
Oncology	0.37
Liquid	0.47
Solid	0.36

Table 3: Inductive 2-layer GCN macro-F1 across the four scopes.

Full vs Solid. On the 623 shared pairs, both Full-trained and Solid-trained RF reach approximately 88% pairwise accuracy; the ~ 0.22 macro-F1 gap visible in Table 1 is therefore *not* driven by per-pair accuracy differences on common pairs, but by *rare-class enrichment* in Solid’s full test set — a consequence of the scope-specific label distribution that the shared subset does not capture. The gap is real for deployment purposes but is not a per-pair-accuracy effect on shared pairs.

Liquid vs Solid. A substantial pairwise-accuracy gap ($\Delta = -0.130$) but a non-significant p -value, with only $n = 23$ shared pairs available after intersection. This comparison is *underpowered*, and the appropriate conclusion is *insufficient evidence*, not *no difference*.

5.3 Mechanism: RF vs GCN

The GCN’s macro-F1 is markedly lower than RF across all scopes, and is *non-monotone* in scope (Table 3).

The pattern is qualitatively different from RF. The GCN does not benefit from larger scope, and the non-monotonicity (Liquid > Full) rules out a simple scale-effect interpretation.

A second observation localizes the source of the RF scope effect. Across every scope, the two strongest predictors by RF feature-importance ranking are *common neighbors* and *Jaccard similarity*. These are precisely the features that encode neighborhood overlap, and whose informativeness is, by construction, sensitive to neighborhood density.

We treat the contrast between Tables 1 and 3 as the central comparative finding of the paper: the scope effect that supports H_1 is **mechanism-specific** — visible to a model that consumes explicit pairwise features, invisible to a model that propagates information end-to-end through message passing. We develop the interpretation in Section 6.

6 Discussion

6.1 Label-mix caveat: the scope effect is observational

The RF macro-F1 declines monotonically from full to solid (Table 1), and McNemar’s tests on shared pairs — confirmed by permutation tests — establish two of three full-vs-specialty contrasts (Table 2). We are confident that the directional claim of H_1 holds for RF on this dataset. We are *not* claiming a clean causal effect of scope on predictability, however, and the reason is label distribution shift.

The seven interaction types are not uniformly distributed across drug subsets. In the full graph, *adverse_effects* accounts for $\sim 32\%$ of edges and *concentration* for $\sim 18\%$. In the oncology subgraph this ordering inverts: *adverse_effects* drops to $\sim 10\%$ while *concentration* rises to $\sim 43\%$. The shift is biologically plausible — oncology drugs are heavily metabolized through CYP3A4 and CYP2D6 and are commonly co-administered with antiemetics, antifungals, and PPIs that themselves modulate CYP activity, generating proportionally more concentration-change interactions. An adverse-effects interaction such as additive QT prolongation is mechanistically generic and arises across drug classes; a concentration interaction is metabolism-specific and concentrates in metabolically active classes.

The consequence is that the observed scope effect on macro-F1 is partly a *label-mix effect*: macro-F1 weights each class equally, and the label mix that the model sees varies systematically with scope. A more careful version of the claim is therefore:

*In this study, full-graph RF predicts DDI types significantly better than oncology-subgraph RF. This effect likely reflects **both** broader network context **and** a shifted label distribution; the current design does not fully disentangle the two.*

We view this explicit caveat as part of the hypothesis test, not a weakness of it. The observational claim is, we think, the honest one for a deployment-relevant question, since a deployed specialty system would also see the natural specialty label distribution.

6.2 Mechanism interpretation: why scope matters for RF, not for GCN

Two observations point to a unified mechanism.

Observation 1 — pairwise features dominate RF. Common neighbors and Jaccard similarity are the strongest predictors across every scope. Both directly encode neighborhood overlap: when two drugs share many neighbors in a dense network, that overlap signals shared targets, shared metabolic pathways, or shared indications. In a sparse subgraph the same statistic is noisy, and the smaller graph simply provides

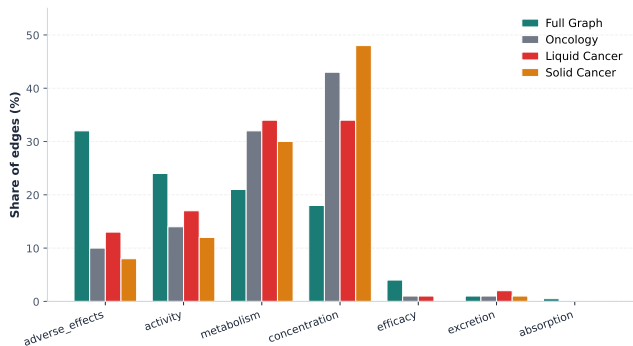


Figure 4: Distribution of interaction types across the four scopes. The shift is most pronounced in the two dominant types: *adverse_effects* drops from 32% in the full graph to 8–13% across oncology scopes, while *concentration* rises from 18% to 34–48%. This is the label-distribution shift that confounds the scope effect on macro-F1 (Section 6).

less of it. This is exactly the dependence on neighborhood density highlighted by the network-medicine literature [7, 8].

Observation 2 — GCN is relatively insensitive to scope. The GCN’s performance does not follow the RF downward trend. A plausible mechanism is dense-graph *over-smoothing*: with average degree ~ 225 , two layers of graph convolution mix neighborhood information aggressively across the network, pushing node representations toward a common mean and reducing class separability [11]. Under this mechanism the GCN’s bottleneck lies in representation collapse, not in scope-related signal availability — so changing the scope does not change the model’s binding constraint, and macro-F1 stays roughly flat (and low).

Synthesis. The scope effect appears strongest in models that **directly exploit explicit pairwise structural features**, especially neighborhood-overlap features that scale informativeness with neighborhood density. A GCN with one-hop or two-hop message passing on a dense graph cannot read off these features as such, and so it cannot capitalize on the scope-related improvement in their quality.

Design implication. A specialty DDI tool built on subgraph data should not rely heavily on neighborhood-overlap features as engineered inputs — they will be unreliable in sparse regions of the graph. It should instead use either drug-intrinsic features (molecular structure, target overlap, pharmacological class) that do not depend on neighborhood density, or representations transferred from full-graph pretraining. Improving the GCN by adding richer initialization or by using architectures less prone to over-smoothing (e.g. attention-based or

Condition	macro-F1
Random baseline	0.143
LLM-only (zero-shot)	0.219
LLM + raw graph features (text)	0.232
LLM + retrieved cases (RAG, $K = 5$)	0.839

Table 4: LLM macro-F1 on a sampled test subset ($n = 300$), GPT-4o-mini in all conditions.

residual GNNs) is, we suspect, a more productive direction than scaling depth or width on this corpus.

7 Extension: LLM/RAG follow-up

We additionally ran a separate follow-up study testing whether LLMs can solve the same DDI type prediction task — and how the way graph information is presented changes the answer. This section reports methods and results for that follow-up; the main scope-effect contribution stands independently of these results.

Conditions. We evaluate three configurations of GPT-4o-mini on a sampled test subset ($n = 300$, due to API cost):

- **LLM-only.** Drug names only, zero-shot prompt with brief one-line definitions of each of the seven types.
- **LLM + raw features (text).** The 14 graph features are passed in the prompt as a labeled text vector. Implementation uses LangGraph [16] as the agent framework, but the substantive change relative to LLM-only is the input *representation*: the model now receives structured numerical features alongside drug names.
- **LLM + retrieved cases (RAG, $K = 5$).** For each test pair (u, v) , the $K = 5$ most similar training pairs by cosine similarity over the 14-dimensional graph-feature vector are retrieved, their labels are formatted as in-context analogues, and the model is asked to predict the type of the query pair [15].

Results. Table 4 reports the three conditions alongside the random baseline. Figure 5 visualizes the same numbers as a bar chart.

Interpretation: representation, not information. The intermediate condition — passing the same 14 graph features that drive RF to 0.926, this time as text in the LLM prompt — improves macro-F1 only marginally over zero-shot (0.232 vs 0.219). Both remain near the random baseline and far below feature-based RF. The natural reading is that the LLM cannot consume numerical graph features in the way a tree-based model can: the features are present in the prompt, but

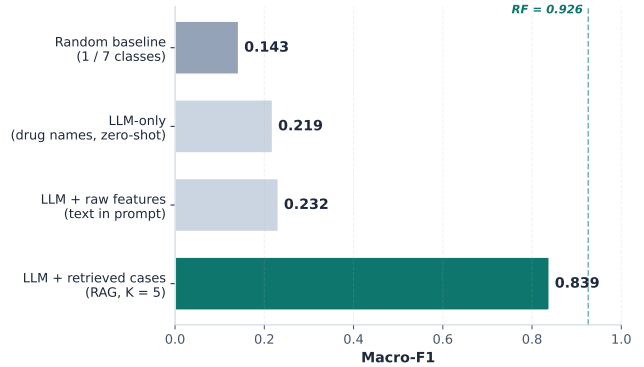


Figure 5: LLM macro-F1 across the three conditions, with random baseline ($1/7 = 0.143$) and feature-based RF reference (0.926, dashed line). The same model with comparable amounts of in-context information moves from 0.232 (raw features as text) to 0.839 (retrieved labeled analogues).

they are not in a representational form the pretrained model can exploit. A 14-dimensional vector of structural statistics is not the kind of input on which language models are trained to reason.

The retrieval condition supports this interpretation. With $K = 5$ retrieved similar pairs and their labels included as in-context labeled analogues, the same model reaches macro-F1 = 0.839. The information available to the model is in some sense *less* (no explicit pairwise features), but the *representation* is more useful: labeled analogues are exactly the input modality on which pretrained LLMs are well calibrated. Same model, same task, comparable amounts of in-context information, dramatically different macro-F1: 0.232 vs 0.839.

We label this finding “**representation, not information**” and view it as a stand-alone contribution from the follow-up, separate from the main scope-effect result. Concretely, it implies that LLM-based DDI systems should be built on retrieval-augmented analogue reasoning rather than on prompt-injected numerical feature vectors.

8 Conclusion and future work

We tested one focused hypothesis — that full-network graph features outperform oncology-subgraph features for DDI type prediction — and found:

- **H_1 is supported.** Full-graph RF significantly outperforms oncology RF ($p = 0.036$) and liquid-cancer RF ($p = 0.0003$) on shared test pairs, with both McNemar’s and permutation tests in agreement.
- **Two comparisons need careful framing.** Full vs Solid shows a large macro-F1 gap that does not survive a per-pair test on shared pairs; the gap is explained by rare-class enrichment in Solid’s full test set. Liquid vs Solid ($n = 23$ shared pairs) is underpowered.
- **The scope effect is observational, not causal.**

Label distributions shift systematically between scopes, and the current design does not fully disentangle scope from label mix.

- **The scope effect is mechanism-specific.** A 2-layer GCN is non-monotone in scope (0.36–0.47); the scope effect appears strongest in models that directly exploit explicit pairwise features, especially common neighbors and Jaccard similarity.
- **LLM follow-up: representation, not information.** Drug-names-only zero-shot reaches 0.219; raw features supplied as text barely move the needle to 0.232; retrieval-augmented analogues at $K = 5$ reach 0.839 with the same model.

The practical implication is that, in this DDI setting, graph scope is not just a preprocessing choice — it materially affects predictive performance. Specialty DDI tools should not be built by naively recomputing the same feature pipeline on a specialty subgraph and expecting equivalent performance.

Limitations. The dataset is DrugBank-derived and single-source; the regex extraction is high-coverage but template-bound; the GCN baseline uses minimal initialization and a fixed 2-layer architecture; the LLM follow-up runs on a sampled $n = 300$; the four McNemar contrasts are reported uncorrected for multiple comparisons (under Bonferroni at $\alpha/4 = 0.0125$, Full vs Liquid retains significance while Full vs Oncology does not).

Future work. Five directions follow naturally from these results. *Cross-dataset replication* on TWOSIDES would test whether the scope effect and label-mix confound generalize beyond DrugBank. A *controlled experiment* fixing label distribution across scopes would disentangle scope from label mix, at the cost of discarding most oncology data. *Stronger GCN baselines* — with molecular-feature initialization, attention-based aggregation, or residual connections — would test whether the GCN’s scope-insensitivity is intrinsic or an artifact of our minimal architecture. *Frontier-scale LLM testing* (GPT-4o, Claude 4) would probe whether the representation-vs-information bottleneck persists at larger model scale. Finally, *specialty-curated case bases* for retrieval-augmented inference appear to be the most promising near-term route to deployable specialty DDI tools.

Code and data availability

Code, notebooks, and extracted features are available at <https://github.com/ewu18/ddi-graph-llm>. The dataset is publicly available on Kaggle (DrugBank-derived).

References

- [1] S. Vilar, E. Uriarte, L. Santana, et al. Similarity-based modeling in large-scale prediction of drug-drug interactions. *Nature Protocols* 9(9):2147–2163, 2014.
- [2] J. Y. Ryu, H. U. Kim, S. Y. Lee. Deep learning improves prediction of drug-drug and drug-food interactions. *PNAS* 115(18):E4304–E4311, 2018.
- [3] M. Zitnik, M. Agrawal, J. Leskovec. Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics* 34(13):i457–i466, 2018.
- [4] D. S. Wishart, Y. D. Feunang, A. C. Guo, et al. DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Research* 46(D1):D1074–D1082, 2018.
- [5] S. Vashishth, S. Sanyal, V. Nitin, P. Talukdar. Composition-based multi-relational graph convolutional networks. *ICLR*, 2020.
- [6] D. Liben-Nowell, J. Kleinberg. The link-prediction problem for social networks. *Journal of the American Society for Information Science and Technology* 58(7):1019–1031, 2007.
- [7] A.-L. Barabási, N. Gulbahce, J. Loscalzo. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics* 12(1):56–68, 2011.
- [8] I. A. Kovács, K. Luck, K. Spirohn, et al. Network-based prediction of protein interactions. *Nature Communications* 10(1):1240, 2019.
- [9] T. N. Kipf, M. Welling. Semi-supervised classification with graph convolutional networks. *ICLR*, 2017.
- [10] W. Hamilton, Z. Ying, J. Leskovec. Inductive representation learning on large graphs. *NeurIPS*, 2017.
- [11] Q. Li, Z. Han, X.-M. Wu. Deeper insights into graph convolutional networks for semi-supervised learning. *AAAI*, 2018.
- [12] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, E. Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, P10008, 2008.
- [13] Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12(2):153–157, 1947.
- [14] B. Efron. Bootstrap methods: another look at the jack-knife. *Annals of Statistics* 7(1):1–26, 1979.
- [15] P. Lewis, E. Perez, A. Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.
- [16] LangChain Inc. LangGraph: a library for building stateful, multi-actor applications with LLMs. <https://github.com/langchain-ai/langgraph>, 2024.